Parameters of Microring modulators.

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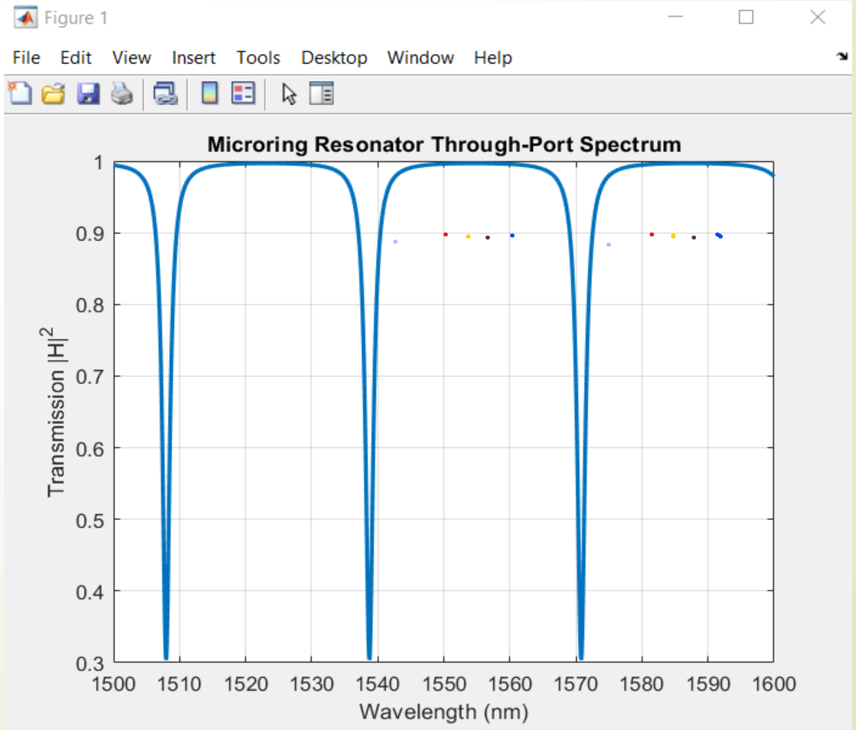
- FSR
- Q-factor
- Finess
- Δn

Microring modulators

Free Spectral Range (FSR)

- The separation of successive resonances is termed the FSR.
- In WDM systems, multiple optical channels are carried simultaneously, each at a different wavelength.



Microring modulators

Free Spectral Range (FSR)

FSR should be **equal to or larger than** your total operational bandwidth. **For N channels with spacing Δλ:**

$$\text{FSR} \geq N \times \Delta\lambda$$

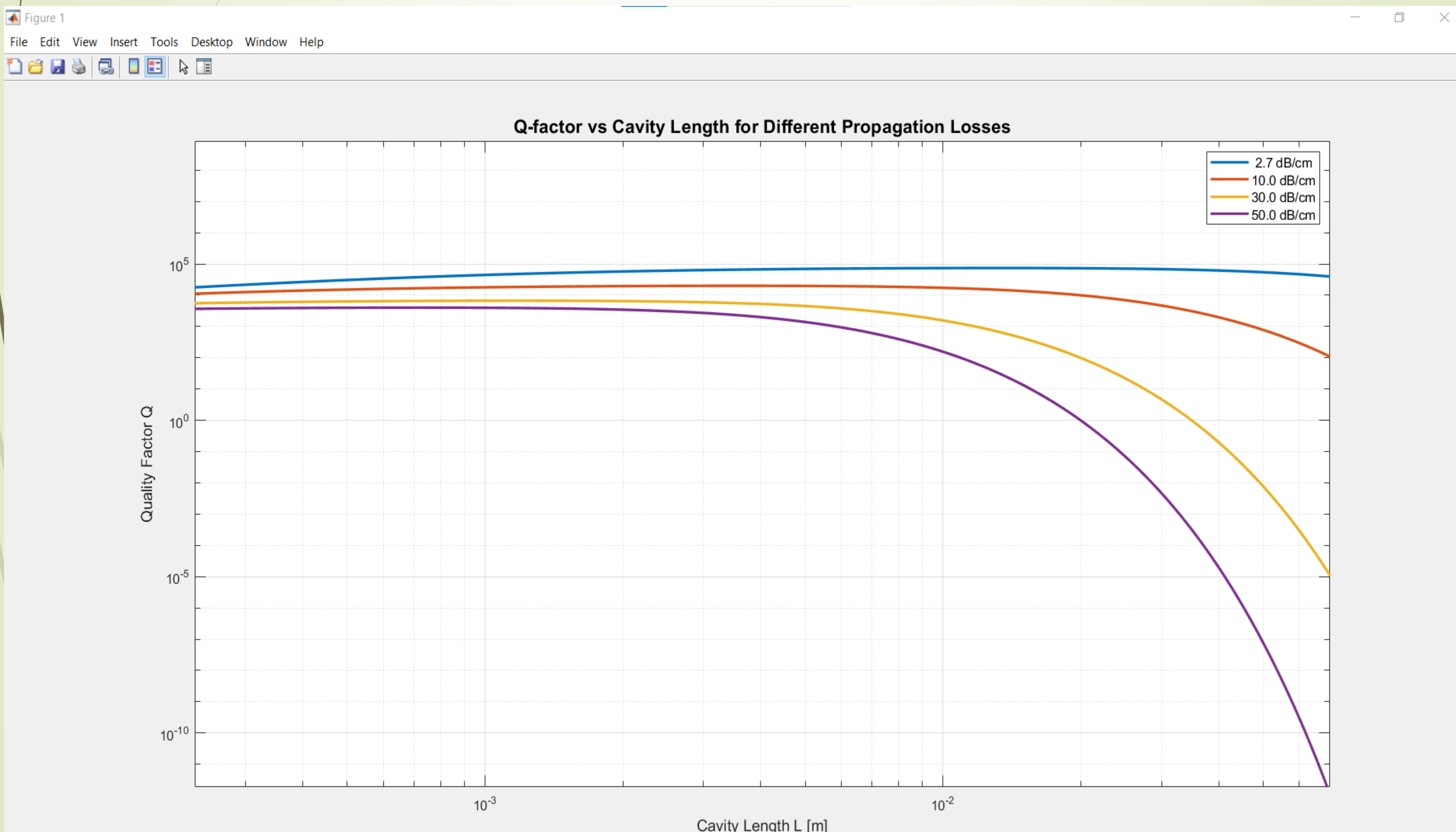
Example: For 40-channel DWDM with 100 GHz spacing:
Channel spacing: 100 GHz = 0.8 nm @ 1550 nm
Total bandwidth: 40 × 100 GHz = 4 THz ≈ 32 nm

$$\text{FSR}_{\lambda} = \frac{\lambda^2}{ngL}$$

$$\frac{df}{d\lambda} = \frac{c}{\lambda^2}$$

Microring modulators

Q-factor



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Q-factor $\quad$ $Q = \frac{\lambda}{FWHM}$

-With propagation losses set to 2.7 dB/cm, the highest Q-factor that can be obtained under the given conditions is about 1.42 $*10^5$ with an APF resonator of approximately 10 mm roundtrip length.

- It is a measure of the temporal confinement (**The Q factor is proportional to the photon lifetime in the cavity**) of light inside the ring resonator and how sharp resonance is.
- It measures the wavelength selectivity of the ring resonator.
- Increasing L reduces FWHM and thus increases Q.
- At short cavities total loss is dominated by **fixed coupling losses**, but at longer cavities Propagation losses dominate: Loss $\propto e^{-\alpha L}$

Microring modulators

Finesse

$$F = \frac{FSR}{FWHM}$$

-**Finesse (F)** physically represents how many times a photon can circulate in the resonator before being lost.

It indicates number of round-trips a photon makes before escaping.

-Finesse decreases with increasing cavity length, as FSR decreases as cavity length increases.

-also longer paths accumulated more loss per round-trip, reducing the number of effective circulations each photon could make before being lost.

Microring modulators

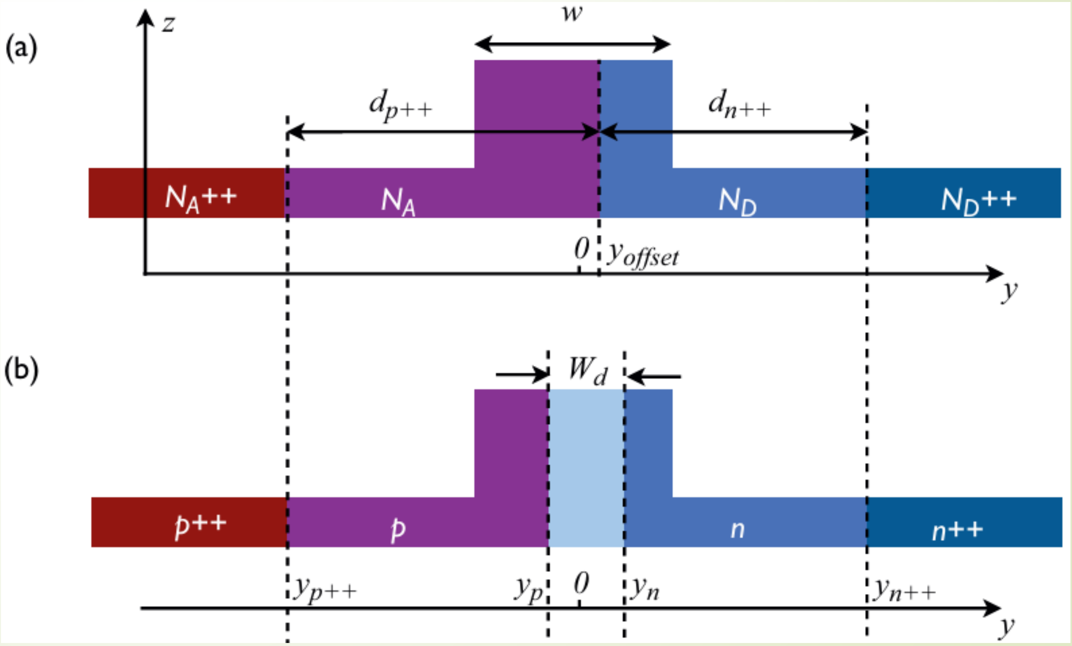
THE MEASURED FSR AND FINESSE FOR DIFFERENT DEVICES

Index difference	Device Diameter	FSR (nm) 1300nm	Finesse 1300nm	Loss* (dB/cm)	FSR 1550nm	Finesse 1550nm	Loss* (dB/cm)
0.3	64μm	5	141	9.03	7.5	117	7.5
0.3	55μm	5.4	92	16.58	8.1	84	14.7
0.1	440μm	0.8	80	1.93			
0.1	660μm				0.8	20	2.56

* Excluding material loss

Microring modulators

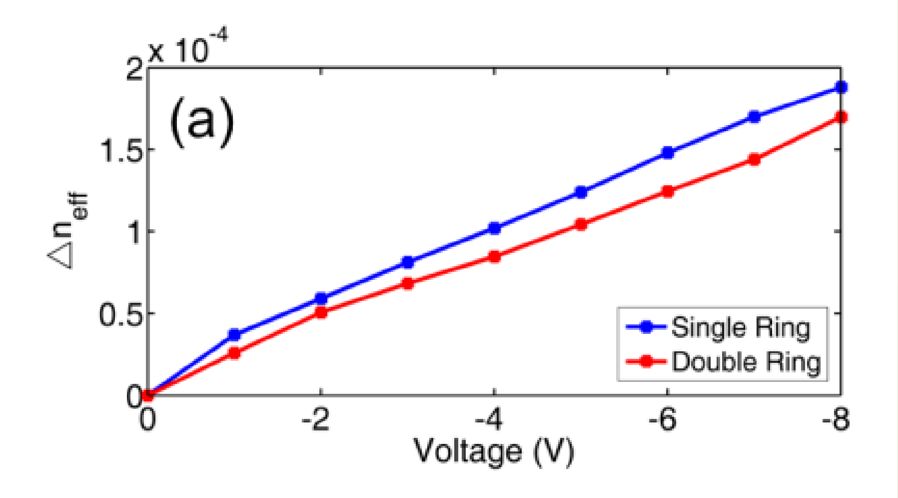
- the concentration of carriers is varied either by injecting or removing carriers from the device (This is termed the plasma dispersion effect).
- Controlling the width of the depletion region is done by applying external voltage.
- Reversed->depletion
  Forword->injection
- Thus, the Concentration variation changes the refractive index and absorption.



Microring modulators

$$\Delta\alpha = \Delta\alpha_e + \Delta\alpha_h = 8.88 \times 10^{-21}(\Delta N_e)^{1.167} + 5.84 \times 10^{-20}(\Delta N_h)^{1.109}$$

$$\Delta n = \Delta n_e + \Delta n_h = -[5.40 \times 10^{-22}(\Delta N_e)^{1.011} + 1.53 \times 10^{-18}(\Delta N_h)^{0.838}]$$



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